# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — MATHS UNIT TEST SET C

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** The expression $a^2 + 2ab + b^2$ is the expansion of: 1

**(a)** $a^2 - b^2$ **(b)** $(a+b)(a-b)$ **(c)** $(a+b)^2$ **(d)** $(a-b)^2$

**2.** The value of $102 \times 15$ is: 1

**(a)** 1530 **(b)** 1515 **(c)** 1540 **(d)** 1545

**3.** ₹ 720 is divided between A and B in the ratio 5 : 4. A's share is: 1

**(a)** ₹ 360 **(b)** ₹ 320 **(c)** ₹ 480 **(d)** ₹ 400

**4.** The ratios 2 : 5 and 6 : x are equal if x is: 1

**(a)** 15 **(b)** 18 **(c)** 12 **(d)** 30

**5.** $\frac{7}{20}$ expressed as a percentage is: 1

**(a)** 70% **(b)** 30% **(c)** 40% **(d)** 35%

**6.** The simple interest on ₹ 4000 at 7.5% per annum for 2 year(s) is: 1

**(a)** ₹ 300 **(b)** ₹ 4600 **(c)** ₹ 1200 **(d)** ₹ 600

**7.** A quantity of 1500 is decreased by 12%. The new value is: 1

**(a)** 1500 **(b)** 1320 **(c)** 180 **(d)** 1680

**8.** The hypotenuse of a right triangle is 5 cm and one leg is 3 cm. The other leg is: 1

**(a)** 3 cm **(b)** 2 cm **(c)** 4 cm **(d)** 5 cm

**9.** Which is the longest side of a right triangle? 1

**(a)** hypotenuse **(b)** base **(c)** median **(d)** altitude

**10.** The Baudhayana–Pythagoras theorem states that in a right triangle, the square on the hypotenuse equals: 1

**(a)** the sum of the legs
**(b)** the difference of the squares
**(c)** the sum of squares on the other two sides
**(d)** twice the product of the legs

**11.** In a pie chart, a sector representing 40% of the data has a central angle of: 1

**(a)** 40° **(b)** 144° **(c)** 80° **(d)** 154°

**12.** A whole divided in the ratio 3 : 7 gives the smaller part as what fraction of the whole? 1

**(a)** 7/10 **(b)** 3/4 **(c)** 3/7 **(d)** 3/10

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 10–15 words each.*

**13.** Find $63 \times 57$ using $(60+3)(60-3)$. 2

**14.** What is a ratio? Write the ratio of 24 to 36 in simplest form. 2

**15.** A shirt of marked price ₹ 800 is sold at 15% discount. Find the selling price. 2

**16.** A ladder 13 m long reaches 12 m up a wall. How far is its foot from the wall? 2

**17.** 6 men build a wall in 12 days. How long will 9 men take? 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 30–40 words each.*

**18.** Evaluate $97^2 - 3^2$ without calculating each square. 3

**19.** Two numbers are in the ratio 3 : 8 and their sum is 66. Find the numbers. 3

**20.** A shopkeeper allows successive discounts of 20% and 10% on a marked price of ₹ 2000. Find the selling price. 3

**21.** On a map, 1 cm stands for 25 km. Two cities are 12.4 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

**22.** Find $99^2$ using a suitable identity. 3

OR

Mind the mistake: a student writes $(x+5)^2 = x^2 + 25$. Find the mistake and correct it. 3

**23.** Divide ₹ 2800 in the ratio 3:4:7. 3

OR

A mixture of 40 L contains milk and water in the ratio 7 : 1. How much water must be added to make the ratio 7 : 3? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–100 words each.*

**24.** **Case study:** A recipe for 6 people uses 450 g flour, 300 g sugar and 150 g butter. 5

(i) Write the ratio of flour : sugar : butter.
(ii) Find quantities for 10 people.
(iii) How many people can be served with 1.5 kg flour?

OR

(i) Define ratio and proportion. 5
(ii) Find $x$ if $x : 12 = 15 : 20$.
(iii) Check if 6, 9, 10, 15 are in proportion.

**25.** (i) Find the length of the altitude of an equilateral triangle of side 10 cm. 5
(ii) Hence find its area.

OR

(i) Prove that the triangle with sides 20, 99, 101 is right-angled. 5
(ii) Find its area.

## END OF QUESTION PAPER — Total: 50 Marks